

Scenario S2-O3-I1 — Our Level 3 (Fully Funded) / Investor Level 1 (Bootstrapped)

Scenario Summary

Structurally contradictory path. The founders self-fund at fully funded intensity (~150% of ideal spend) through Pre-Pilot and Pilot 1, deploying corporate-level resources for 4-6 months, then a bootstrapped-level investor arrives during the Pilot 1 to Pilot 2 transition (approximately M5-M7). The mismatch is stark: the company builds momentum with high-intensity resourcing, then downshifts to a bootstrapped investor who cannot sustain that pace. Operations slow, team utilization drops, and the momentum gained in early phases is partially wasted.

- **Economic cost to Production:** ~\$163,000 / INR 13,529,000
- **Target timeline to Production:** 26 months
- **Founder/company capital before investor:** ~\$124,000 / INR 10,292,000
- **Investor capital modeled:** ~\$39,000 / INR 3,237,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Main risk to watch:** momentum collapse at Pilot 2 when investor capital is insufficient to sustain the operational tempo established during the founder-funded phases

Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M4	4 months	Architecture freeze (M1), APK backlog lock (M2), doctor training draft (M3), admin callback SOP (M3), internal rehearsal cycles (M4).	Core web flows stable at full scope, internal rehearsals pass, doctor and admin runbooks versioned. Faster than L1/L2 due to full team utilization.
Pilot 1	M4-M6	2 months	Warm-network onboarding, first paid consults, callback calibration, doctor playbook iteration, investor pitch deck finalized.	Ethiopia pilot runs live for 2 months with measured throughput; investor term sheet signed.
Pilot 2	M6-M14	8 months	Kenya prep (M10), partner reporting (M11), roster planning, partial automation, staffing calibration, refresher training. Pace slows significantly from Pre-Pilot due to bootstrapped investor capital.	5k-10k user-capable operations with APK improvements, reporting, and staffed callback coverage — achieved more slowly than the early pace would suggest.
Production Readiness	M14-M26	12 months	Compliance closeout (M18), release freeze (M21), launch playbooks (M23), certification refreshers, production support schedule.	Current prototype scope is production-safe across web and patient mobile with documented fallback operations. Extended timeline due to under-funded later phases.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at full pace; APK build runs in parallel with web work; doctor/admin training materials drafted; support desk scripts and callback trees defined. Full team utilization enables broad feature coverage in compressed timeline.
Pilot 1	Patient app covers core journeys (registration, dashboard, basic consult); provider web is stable for GP and specialist flows; doctor roster, admin callbacks, and manual fulfilment visible in-system. Broader feature set than L1/L2 due to Pre-Pilot intensity.
Pilot 2	Core patient features complete but iteration slows; partner dashboards functional; staffing ladder active at minimum levels; reporting and payment loops operational but progress decelerates as bootstrapped investor capital constrains ops and infra spend. Kenya prep begins late in phase.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Extended timeline required because bootstrapped investor capital stretches thin across 12 months of production hardening.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	96.0%	Next business day	3 core team at full utilization + 7 part-time developers at ~95% capacity + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	97.0%	Same-day for critical issues	Minimum employed coverage: 3 doctors, 3 admin ops, 2 technical support. Strong early ops presence funded by founder capital.
Pilot 2	97.5%	4-hour P1 response	Reduced employed coverage: 3 doctors, 3 admin ops, 2 technical support. Bootstrapped investor cannot fund the ladder increase. Flat from Pilot 1.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 5 doctors, 5 admin ops, 3 technical support. Below ideal but manageable. Constrained by remaining investor capital.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	\$86,700 / INR 7,196,100
Operating expenditure before Production launch	\$76,300 / INR 6,332,900
Total economic spend to Production	\$163,000 / INR 13,529,000

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	\$68,640	INR 5,697,120	3 core team at \$20/hr x 88 hrs/mo x 95% utilization for M0-M6 (\$5,016/mo), dropping to 50% post-investor (\$2,640/mo) for M7-M26.
Web platform hardening	\$14,900	INR 1,236,700	745 hours at \$20/hr senior rate. Compressed into early phases due to full team availability.
Patient APK build	\$7,200	INR 597,600	Fixed Android quote: 480 hours at \$15/hr. Runs in parallel with web work during Pre-Pilot.
QA, DevOps, security	\$8,800	INR 730,400	~440 hours at \$20/hr. Full hardening in Pre-Pilot; maintenance-level post-investor.
Admin and care operations	\$78,960	INR 6,553,680	Employed doctors, admin callback staff, and technical support ladder across 22 months of live operations. See ops staffing breakdown below. Constrained growth post-investor.
Partner onboarding and provider success	\$10,500	INR 871,500	Doctor/admin training protocols, SOP writing, partner reporting. Well-resourced in Pre-Pilot, lean post-investor.
Legal, accounting, compliance	\$5,500	INR 456,500	Friendly-network pricing at economic value. Slightly higher than L1 due to more thorough early-phase compliance work.
Infrastructure, phones, and tools	\$22,500	INR 1,867,500	Hosting, devices, telecom/call costs, operational tooling across 26 months. Starts high, drops post-investor.
Marketing, travel, rehearsals	\$10,000	INR 830,000	Moderate spend. Warm-network and targeted digital in early phases, then constrained.
Contingency and buffer	\$36,000	INR 2,988,000	~22% buffer on total. Lower than S2-O1-I1 because early phases are well-funded; risk concentrates in later phases.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M4)	4 months	\$0	INR 0	\$0
Pilot 1 (M4-M6)	2 months	\$3,723	INR 308,990	\$7,446
Pilot 2 (M6-M14)	8 months	\$3,723	INR 308,990	\$29,784
Production Readiness (M14-M26)	12 months	\$5,144	INR 426,952	\$61,728
Total operational staffing				\$98,958

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	4 months	\$130	INR 10,790	\$520
Pilot 1	2 months	\$450	INR 37,350	\$900
Pilot 2	8 months	\$400	INR 33,200	\$3,200
Production Readiness	12 months	\$700	INR 58,100	\$8,400
One-time equipment (phones, laptops)	—	—	—	\$1,500
Total infrastructure				\$14,520

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~95% effective utilization pre-investor	Single owner/founder + 2 founding members. Full parallelism of tasks during early phases.
Technical delivery (pre-investor)	7 part-time developers, modeled at ~640 productive hrs/mo	~150% of ideal 480 hrs/mo. Maximum utilization with overtime/weekend sprints.
Technical delivery (post-investor)	7 part-time developers, modeled at ~240 productive hrs/mo	Bootstrapped investor capital forces sharp reduction to ~50% utilization. Significant deceleration.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 3 doctors / Pilot 2: 3 / Production: 5	Growth stalls in Pilot 2 due to investor capital constraints.
Employed admin ops	Pilot 1: 3 admins / Pilot 2: 3 / Production: 5	Callbacks, exception handling, manual fulfilment. Flat through Pilot 2.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M6)	~\$22,245	~INR 1,846,335	~150% of full \$14,830/mo rate card. Full corporate investment.

Post-investor (M7-M26)	~\$7,400	~INR 614,200	~50% of full rate card. Bootstrapped investor cannot sustain the early pace.
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Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~\$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Early availability due to Pre-Pilot intensity	Grows slowly; constrained by ops staffing plateau	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	\$3.5 / ETB 470	\$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by mid-Production.
Specialist consult	\$9 / ETB 1,250	\$12 / KES 1,560	Available from late Pilot 1 due to full Pre-Pilot build.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	\$10-15	\$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	\$8	\$14	\$20
Gross margin	40%	45%	55%
Monthly ARPU	\$1.8	\$2.5	\$4
Monthly churn	13%	11%	8%
LTV/CAC target	1.5x	2.0x	2.8x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M4)	\$18,200	INR 1,510,600	0.8 months	Critical: high burn consumes founder capital quickly. Investor must be in pipeline by M3.
Pilot 1 (M6)	\$11,400	INR 946,200	0.4 months	Extremely tight. Investor close must happen by M7 or operations collapse.
Pilot 2 (M14)	\$25,800	INR 2,141,400	2.4 months	Trigger contingency if cash drops below 2 months of burn.
Production Readiness (M26)	\$18,500	INR 1,535,500	2.1 months	Trigger contingency if cash drops below 2 months of burn. Revenue must ramp to sustain.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	\$22,800	\$0	\$0	\$22,375	\$425	\$425
M2	Pre-Pilot	\$22,800	\$0	\$0	\$22,375	\$425	\$850
M3	Pre-Pilot	\$22,800	\$0	\$0	\$22,375	\$425	\$1,275
M4	Pre-Pilot	\$22,800	\$0	\$0	\$22,375	\$425	\$1,700
M5	Pilot 1	\$22,800	\$0	\$500	\$26,548	-\$3,248	-\$1,548
M6	Pilot 1	\$22,800	\$0	\$1,000	\$26,548	-\$2,748	-\$4,296
M7	Pilot 2	\$0	\$39,000	\$1,800	\$11,523	\$29,277	\$24,981
M8	Pilot 2	\$0	\$0	\$2,200	\$11,523	-\$9,323	\$15,658
M9	Pilot 2	\$0	\$0	\$2,600	\$11,523	-\$8,923	\$6,735
M10	Pilot 2	\$0	\$0	\$3,000	\$11,523	-\$8,523	-\$1,788
M11	Pilot 2	\$0	\$0	\$3,400	\$11,523	-\$8,123	-\$9,911

M12	Pilot 2	\$0	\$0	\$3,800	\$11,523	-\$7,723	-\$17,634
M13	Pilot 2	\$0	\$0	\$4,200	\$11,523	-\$7,323	-\$24,957
M14	Pilot 2	\$0	\$0	\$4,600	\$11,523	-\$6,923	-\$31,880
M15	Prod. Readiness	\$0	\$0	\$5,200	\$12,844	-\$7,644	-\$39,524
M16	Prod. Readiness	\$0	\$0	\$5,800	\$12,844	-\$7,044	-\$46,568
M17	Prod. Readiness	\$0	\$0	\$6,500	\$12,844	-\$6,344	-\$52,912
M18	Prod. Readiness	\$0	\$0	\$7,500	\$12,844	-\$5,344	-\$58,256
M19	Prod. Readiness	\$0	\$0	\$8,500	\$12,844	-\$4,344	-\$62,600
M20	Prod. Readiness	\$0	\$0	\$9,500	\$12,844	-\$3,344	-\$65,944
M21	Prod. Readiness	\$0	\$0	\$10,500	\$12,844	-\$2,344	-\$68,288
M22	Prod. Readiness	\$0	\$0	\$11,500	\$12,844	-\$1,344	-\$69,632
M23	Prod. Readiness	\$0	\$0	\$13,000	\$12,844	\$156	-\$69,476
M24	Prod. Readiness	\$0	\$0	\$14,500	\$12,844	\$1,656	-\$67,820
M25	Prod. Readiness	\$0	\$0	\$16,000	\$12,844	\$3,156	-\$64,664
M26	Prod. Readiness	\$0	\$0	\$18,000	\$12,844	\$5,156	-\$59,508

Totals:

- Founder capital in: \$136,800 / INR 11,354,400 (M1-M6 at ~\$22,800 x 6 months)
- Investor capital in: \$39,000 / INR 3,237,000
- Cumulative revenue: ~\$153,500
- Cumulative outflow: ~\$385,200

Critical note: The cash flow table reveals the fundamental flaw of this scenario. The company goes cash-negative by M10 and remains negative through the end of the planning horizon. The \$39,000 investor capital is wholly insufficient to bridge Pilot 2 and Production. This scenario requires either (a) additional founder capital injection post-M6, (b) an emergency bridge loan during Pilot 2, or (c) revenue ramp far above what is modeled. Without one of these interventions, the company runs out of cash.

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	\$175,800 / INR 14,591,400
Total economic cost	\$163,000 / INR 13,529,000
Sweat-equity gap (economic cost minus cash invested)	Negative: cash exceeds economic cost because the scenario is structurally underfunded post-investor
Cumulative revenue at M26	~\$153,500 / INR 12,740,500
Monthly break-even (revenue >= outflow)	~M23 (month 23)
Cumulative break-even (total revenue >= total capital)	~M33-M36 (7-10 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M38-M42 (12-16 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	\$136,800 / INR 11,354,400
Investor round size	\$39,000 / INR 3,237,000
Recommended structure	SAFE or small convertible note
Modeled pre-money valuation	\$350,000
Modeled post-money valuation	\$389,000
Modeled investor dilution from the 20% pool	10.0%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	57.0%	Retained strong majority control.
Founding + strategic pool	20%	19.0%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	10.0% issued / 10.0% remaining	Small allocation from investor pool. Low dilution but low capital value.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
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Cash depletion by M10-M12	Very High	Critical	Bootstrapped investor capital is insufficient. Pre-negotiate bridge loan or additional founder injection.
Momentum collapse at Pilot 2	High	Critical	Team goes from 150% utilization to 50% overnight. Psychological impact of deceleration. Maintain clear communication about the shift.
Investor adds negligible capital for significant governance overhead	High	High	\$39,000 investor may demand board observer rights, reporting, or consent rights disproportionate to their contribution. Protect governance in term sheet.
Team fatigue from intensity whiplash	High	High	The shift from corporate-level to bootstrapped-level is jarring. Prepare the team for the transition.
Over-built product without matching ops capacity	Medium	High	Pre-Pilot builds features at full speed; Pilot 2 cannot staff the operations to support them. Features sit underutilized.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Count friendly counsel at economic value.
Payment-rail mismatch	Medium	Medium	Use digital invoicing first; localize M-Pesa / Telebirr before scaling.
Kenya market entry delayed	Medium	Medium	Focus on Ethiopia stability first. Kenya prep pushed late due to capital constraints.

Recommendation

S2-O3-I1 is a **structurally contradictory scenario** and should be **avoided**. The core problem is a capital mismatch: the founders invest at corporate intensity (\$22,800/month for 6 months = \$136,800), building significant momentum and operational expectations, then a bootstrapped investor arrives with only \$39,000 — barely enough to cover 3 months of even the reduced burn rate.

This scenario fails for three specific reasons:

- 1. Cash flow is structurally negative.** The company goes cash-negative by M10 and never recovers within the planning horizon without additional capital injections. A \$39,000 investor contribution cannot bridge 20 months of Pilot 2 + Production at any reasonable burn rate.
- 2. Momentum whiplash.** The team operates at 150% intensity for 6 months, building features, onboarding partners, and establishing operational tempo — then abruptly drops to 50% utilization. This is psychologically destructive and operationally wasteful.
- 3. Misaligned investor incentives.** An investor contributing \$39,000 to a company that has already deployed \$136,800 in founder capital has very different risk exposure and return expectations. The governance and reporting overhead of managing this relationship may exceed the capital value.

If the founders have the capital for O3 spending, they should pair with an I2 or I3 investor (see S2-O3-I2 or S2-O3-I3) to maintain momentum through scaling phases. Alternatively, they should reduce their own spend to O2 and pursue S2-O2-I2 for a more balanced trajectory.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at \$15/hr average across the mobile team.
 - Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
 - Salary benchmarks for employed coverage: INR 60,000 for doctors (\$720/mo), INR 18,000 for admin ops (\$217/mo), INR 25,000 for technical support (\$301/mo).
 - Our L3 (fully funded) spend is modeled at ~~150% of ideal team utilization, yielding ~\$22,800/mo founder contribution pre-investor~~ (\$22,245 team + ~\$555 ops/infra/overhead).
 - Investor L1 (bootstrapped) contribution sized at ~\$39,000, representing bare-minimum capital from an investor entering a well-funded founder operation.
 - All INR conversions at 1 USD = 83 INR per params.md.
 - FX buffers of 10-15% applied to ETB and KES revenue projections.
 - The cash flow model reveals structural insolvency; this scenario is documented for completeness and to illustrate why O3-I1 pairings should be avoided.
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End of S2-O3-I1 scenario. All assumptions trace to params.md v2.